

1.问题一的程序

```
clc;clear;
a=4315; % 游梁前臂
b=2495; % 游梁后臂
l=3675; % 连杆长度
r=950; % 曲柄半径
q=0.2585; % 按曲柄滑块机构计算 中的 兰格他
W=0:pi/6:2*pi;%转角
w=7.6*pi/30;%角速度
t=0:(60/7.6/143):(60/7.6);
% 按曲柄滑块机构计算
SA_qb=r*(1-cos(w*t)+1/q*(1-sqrt(1-q.^2*sin(w*t).^2)))*(a/b/1000)
VA_qb=w*r*(sin(W)+q/2*sin(2*W))*(a/b/1000);%速度
WA_qb=w.^2*r*(cos(W)+q*cos(2*W))*(a/b/1000)%加速度
```

%按简谐运动计算

```
%SA_jx=(a/b/1000)*r*(1-cos(W)); %位移
SA_jx=(a/b/1000)*r*(1-cos(w*t)) %位移
VA_jx=(a/b/1000)*w*r*sin(W);%速度
WA_jx=(a/b/1000)*w.^2*r*cos(W) %加速度
```

%实际运算数据

```
load 'D:\MATLAB6p5\work\建模\A.txt';
W=0:30:(length(W)-1)*30;
figure(1)
plot(W,VA_qb,'b');
hold on;
plot(W,VA_jx,'r');
grid on;
```

```
figure(2)
plot(W,WA_qb,'b');
hold on;
plot(W,WA_jx,'r');
grid on;
```

```
figure(3)
grid on;
plot(t,SA_jx,'b');
hold on;
plot(t,SA_qb,'r');
grid on;
```

2 问题二的一级杆的泵功图及问题三两种计算产液量方法的代码

```
clc;clear;
```

```

D=[39.7,39.665,39.665,38.131,31.822,30.184,28.372,27.047,26.385,23.806,19.17,16.765,14.988,13.245,11.885,1
4.221,17.114,19.693,20.808,22.446,23.422,21.087,18.996,16.451,15.545,13.698,13.001,14.465,16.277,18.159,17
.183,17.323,17.288,16.521,15.371,14.709,15.057,14.43,14.256,14.36,14.534,14.709,14.883,14.778,14.813,14.70
9,14.953,15.266,15.406,15.301,15.51,16.242,16.73,17.253,17.671,18.612,19.031,19.344,20.146,20.808,21.784,2
1.959,22.69,23.213,23.492,23.353,23.388,23.806,24.12,24.538,24.886,25.339,28.546,31.16,33.251,35.412,37.95
7,40.78,46.148,50.4,53.99,55.942,56.883,57.092,57.545,58.103,58.138,58.242,58.347,58.068,58.068,58.242,58.6
61,58.905,59.462,60.02,60.891,61.205,61.344,60.891,61.275,61.066,60.752,59.811,59.497,58.87,58.033,57.615,
57.232,56.465,55.524,55.036,54.408,53.955,53.502,53.537,53.049,53.223,52.7,52.247,52.178,52.247,51.481,50.
783,50.33,49.807,49.041,48.274,47.298,46.74,45.834,45.59,45.207,44.963,44.684,44.405,44.196,43.917,43.603,
43.115,42.732,41.965,40.85,40.257];
U=[2.518,2.512,2.504,2.495,2.484,2.472,2.459,2.444,2.427,2.41,2.391,2.372,2.351,2.329,2.305,2.281,2.256,2.22
9,2.201,2.172,2.142,2.111,2.079,2.046,2.011,1.975,1.938,1.9,1.86,1.82,1.778,1.735,1.69,1.645,1.598,1.55,1.501,
1.451,1.399,1.347,1.294,1.239,1.184,1.129,1.072,1.015,0.958,0.901,0.844,0.786,0.729,0.673,0.618,0.563,0.51,0.
459,0.409,0.361,0.315,0.272,0.231,0.193,0.158,0.127,0.098,0.074,0.052,0.034,0.02,0.01,0.003,0,0,0.004,0.012,0.
023,0.037,0.054,0.074,0.097,0.123,0.152,0.183,0.216,0.251,0.289,0.328,0.369,0.412,0.456,0.501,0.548,0.596,0.6
45,0.694,0.745,0.797,0.849,0.901,0.954,1.008,1.062,1.116,1.17,1.225,1.279,1.334,1.388,1.442,1.496,1.55,1.603,
1.655,1.707,1.758,1.808,1.858,1.906,1.953,1.999,2.043,2.086,2.127,2.167,2.205,2.241,2.275,2.307,2.337,2.365,2
.391,2.414,2.435,2.454,2.471,2.485,2.497,2.507,2.515,2.52,2.524,2.525,2.524,2.522];
k=length(D);
% DD=zeros(1,k);
% UU=zeros(1,k);
% for n=73:143
%     DD(mod(n,72))=D(n);
%     UU(mod(n,72))=U(n);
% end
% DD(72)=D(144);
% UU(72)=U(144);
% for n=1:72
%     DD(n+72)=D(n);
%     UU(n+72)=U(n);
% end
% DD=DD*1000; %变为以牛顿为单位
G=25.78*ones(1,k);
DD=(D-G)*1000;
%U=U*(3.2/2.525);
av_n=10; %付氏级数所取项数，本文取 10
%以下为四个付氏系数
dert_o=zeros(1,av_n);
tao_t=zeros(1,av_n);
v=zeros(1,av_n);
q=zeros(1,av_n);
%以下为四个特解系数
aifa_a=zeros(1,av_n);
beita_b=zeros(1,av_n);

```

```

k_k=zeros(1,av_n);
miu_u=zeros(1,av_n);
%以下为载荷和位移值的一部分
F1=zeros(1,k);
U1=zeros(1,k);
%以下为参数
E=2.1e11;           %E 为钢杆弹性模量 Dt=70;
Dt=70;              %Dt 油管内径 mm
Dr=22;              %Dr 抽油杆直径 mm
Ar=(Dr/2000).^2*pi; %Ar 抽油杆截面积 m.^2
u=3e7;              %u 液体粘度, Pa.s
L=792.5;            %L 抽油杆长度 m
pr=8456;            %pr 抽油杆密度 kg/m.^3
w=7.6*pi/30;        %w 为曲柄的角速度
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
a=(E/pr).^(1/2);      %a 为应力波在抽油杆柱中的传播速度
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%以下为求阻尼系数
m=Dt/Dr;
B1=(m.^2-1)/2/log(m)-1;
B2=m.^4-1-(m.^2-1).^2/log(m);
%C=2*pi*u/pr/Ar*(1/log(m)+2/B2*(B1+1)*(B1+2/((w*L/a)/sin(w*L/a)+cos(w*L/a))));
C=0.1;
for n=1:av_n
    for p=1:k
        %以下为四个付氏系数
        dert_o(n)=dert_o(n)+2/k*DD(p)*cos(2*n*pi*p/k);% n=0,1,2,3...,av_n
        tao_t(n)=tao_t(n)+2/k*DD(p)*sin(2*n*pi*p/k); % n=1,2,3...,av_n
        v(n)=v(n)+2/k*U(p)*cos(2*n*pi*p/k);% n=0,1,2,3...,av_n
        q(n)=q(n)+2/k*U(p)*sin(2*n*pi*p/k);% n=1,2,3...,av_n
    end
    %以下为四个特解系数
    aifa_a(n)=n*w/a/sqrt(2)*sqrt(1+sqrt(1+(C/n/w).^2)); %C 为阻尼系数%a 为应力波在在抽油杆柱中的传
播速度
    beita_b(n)=n*w/a/sqrt(2)*sqrt(-1+sqrt(1+(C/n/w).^2)); %
    k_k(n)=(dert_o(n).*aifa_a(n)+tao_t(n).*beita_b(n))/(E*Ar*(aifa_a(n).^2+beita_b(n).^2));%E 为钢杆弹性模
量, Ar 为抽油杆柱的截面积, w 为曲柄的角速度
    miu_u(n)=(dert_o(n).*beita_b(n)+tao_t(n).*aifa_a(n))/(E*Ar*(aifa_a(n).^2+beita_b(n).^2));
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    x=792.5;
    %%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
    O(n)=(k_k(n)*cosh(beita_b(n)*x)+q(n)*sinh(beita_b(n)*x))*sin(aifa_a(n)*x)+(miu_u(n)*sinh(beita_b(n)*x)+v(n)*c
osh(beita_b(n)*x))*cos(aifa_a(n)*x);

```

```
P(n)=(k_k(n)*sinh(beitab(n)*x)+q(n)*cosh(beitab(n)*x))*cos(aifa_a(n)*x)-(miu_u(n)*cosh(beitab(n)*x)+v(n)*sinh(beitab(n)*x))*sin(aifa_a(n)*x);
```

```
dO(n)=(tao_t(n)/E/Ar*sinh(beitab(n)*x)+(q(n)*beitab(n)-v(n)*aifa_a(n))*cosh(beitab(n)*x))*sin(aifa_a(n)*x)+(dert_o(n)/E/Ar*cosh(beitab(n)*x)+(v(n)*beitab(n)+q(n)*aifa_a(n))*sinh(beitab(n)*x))*cos(aifa_a(n)*x);
```

```
dP(n)=(tao_t(n)/E/Ar*cosh(beitab(n)*x)+(q(n)*beitab(n)-v(n)*aifa_a(n))*sinh(beitab(n)*x))*cos(aifa_a(n)*x)-(dert_o(n)/E/Ar*sinh(beitab(n)*x)+(v(n)*beitab(n)+q(n)*aifa_a(n))*cosh(beitab(n)*x))*sin(aifa_a(n)*x);
```

```
end
```

```
    %求初值
```

```
    n=0;
```

```
    dert_o_0=0;
```

```
    v0=0;
```

```
    q0=0;
```

```
    dert_o_0=2/k*sum(DD);
```

```
    v0=2/k*sum(U);
```

```
for N=1:k
```

```
    t=(60/7.6/143)*(N-1);
```

```
    for n=1:10
```

```
        U1(N)=U1(N)+O(n)*cos(n*w*t)+P(n)*sin(n*w*t);
```

```
        F1(N)=F1(N)+dO(n)*cos(n*w*t)+dP(n)*sin(n*w*t);
```

```
    end
```

```
    Uxt(N)=dert_o_0*x/2/E/Ar+v0/2+U1(N);
```

```
    Fxt(N)=(dert_o_0/2+E*Ar*F1(N))/1000;
```

```
end
```

```
%figure
```

```
N=0:(60/7.6/143):(60/7.6);
```

```
figure(1)
```

```
subplot(2,2,1);plot(N,Uxt);
```

```
subplot(2,2,2);plot(N,Fxt);
```

```
subplot(2,2,3);plot(N,U);
```

```
subplot(2,2,4);plot(N,D);
```

```
figure(3)
```

```
plot(Uxt,Fxt,'b');
```

```
hold on ;
```

```
plot(U,D,'r');
```

```
grid on;
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

```
%以下为求有效冲程
```

```
%求曲率
```

```
for i=2:k-1
```

```

P(i)=sqrt((Fxt(i)-Fxt(i-1)).^2+(Uxt(i)-Uxt(i-1)).^2);
P(i-1)=sqrt((Fxt(i-1)-Fxt(i+1)).^2+(Uxt(i-1)-Uxt(i+1)).^2);
P(i+1)=sqrt((Fxt(i+1)-Fxt(i)).^2+(Uxt(i+1)-Uxt(i)).^2);
PP(i)=(P(i)+P(i-1)+P(i+1))/2;
S(i)=sqrt(PP(i)*(PP(i)-P(i))*(PP(i)-P(i-1))*(PP(i)-P(i+1))));
K(i)=2*S(i)/P(i)/P(i-1)/P(i+1);
end
%求曲率变化量
len_K=length(K); %143
deto_K=zeros(1,len_K-1);
for i=1:len_K-1
    deto_K(i)=abs(K(i+1)-K(i));
end
%精确曲率变化量
len_deto_K=length(deto_K); %142
av_deto_K=zeros(1,len_deto_K-4);
for i=3:len_deto_K-2
    av_deto_K(i-2)=(deto_K(i-2)+deto_K(i-1)+deto_K(i)+deto_K(i+1)+deto_K(i+2))/5;
end
%求变化量最大点
IndMin=find(diff(sign(diff(av_deto_K)))>0)+1;
IndMax=find(diff(sign(diff(av_deto_K)))<0)+1;
figure(7);
plot(1:length(av_deto_K),av_deto_K);
hold on;grid on;
plot(IndMax,av_deto_K(IndMax),'k*')

```

N=[15,71,87,141];%四边形的右下角为起始点，逆时针转动，第一个点表示右下角，第二个点表示左下角，第三个左上角，第四个右上角。

```

U_point=zeros(1,4);
D_point=zeros(1,4);
for i=1:4
    n=N(i);
    U_point(i)=Uxt(n);
    D_point(i)=Fxt(n);
end
L_eff=Uxt(15)-Uxt(71)    %有效冲程
figure(5)
plot(U_point,D_point,'r');
hold on;
plot(Uxt,Fxt,'b');
%%%%%%%%%%%%%%
%以下为有效面积法求抽油量

```

%以下为泵功图面积

S_beng=0;

S_li=0;

for i=1:143

width=Uxt(i+1)-Uxt(i);

line=(Fxt(i+1)+Fxt(i))/2;

S_beng=S_beng+width*line;

end

%以下为理论的示功图面积

S_li=(max(D)-min(D))*L_eff;

%有效面积法计算抽油量

qp=1440*pi*(70/1000).^2*3.2*7.6*0.9773/4

K=S_beng/S_li

q_area=K*qp

%有效冲程计算抽油量

q_eff=1440*pi*(70/1000).^2*7.6*997.28*L_eff/4/1000 %换算成吨

3 问题二的多级杆的泵功图及问题三两种计算产液量方法的代码

clc;clear;

U=[0.24,0.19,0.15,0.11,0.08,0.05,0.03,0.01,0.00,0.00,0.01,0.02,0.04,0.07,0.10,0.14,0.18,0.23,0.28,0.34,0.40,0.47,0.53,0.61,0.68,0.76,0.84,0.92,1.01,1.09,1.18,1.27,1.36,1.46,1.55,1.65,1.74,1.84,1.93,2.03,2.12,2.22,2.31,2.41,2.50,2.60,2.69,2.78,2.87,2.96,3.05,3.14,3.22,3.31,3.39,3.47,3.55,3.63,3.71,3.78,3.85,3.92,3.99,4.06,4.12,4.18,4.24,4.30,4.35,4.40,4.45,4.49,4.54,4.58,4.61,4.65,4.68,4.71,4.73,4.76,4.77,4.79,4.79,4.80,4.80,4.80,4.79,4.77,4.75,4.73,4.70,4.67,4.63,4.58,4.53,4.48,4.42,4.35,4.29,4.21,4.14,4.05,3.97,3.89,3.80,3.71,3.61,3.52,3.42,3.32,3.22,3.12,3.01,2.91,2.80,2.70,2.59,2.48,2.37,2.26,2.16,2.05,1.94,1.84,1.73,1.63,1.53,1.43,1.33,1.23,1.14,1.05,0.96,0.87,0.79,0.71,0.63,0.55,0.48,0.42,0.35,0.30];

D=[48.50,48.14,47.96,47.35,46.85,46.99,46.67,46.81,47.46,48.61,50.01,52.84,52.98,53.2,53.56,53.42,52.77,55.46,57.08,59.02,60.67,61.46,62.28,63.07,64.4,67.2,69.78,71.94,74.2,75.21,76.68,76.64,74.42,71.87,70.04,68.67,67.49,67.42,67.45,68.89,70.97,72.58,73.05,73.23,72.87,72.08,70.90,69.96,68.10,67.31,67.63,68.24,68.42,69.17,70.50,71.26,71.47,71.01,70.29,69.53,68.85,68.17,67.99,67.63,67.88,68.6,69.21,69.75,69.78,70.04,70.07,69.64,69.03,68.28,67.77,67.31,67.27,67.38,67.74,68.13,68.6,68.71,68.67,68.17,67.85,67.31,66.81,65.73,65.15,64.72,63.61,63.25,62.82,62.57,62.46,62.68,62.50,61.71,60.92,59.45,58.19,56.97,55.10,53.59,51.55,49.83,47.46,44.73,42.25,39.63,38.98,40.6,43.44,45.48,47.24,48.78,49.68,49.75,49.07,47.31,44.73,43.22,42.43,42.29,42.50,43.47,45.55,46.13,47.74,48.25,48.07,47.53,46.49,45.27,44.58,43.76,43.65,44.05,44.87,46.06,46.67,48.14,48.53];

k=length(D);

G1=8456*(25/2000).^2*pi*523.61*9.8/1000*ones(1,k);

G2=8456*(22/2000).^2*pi*664.32*9.8/1000*ones(1,k);

G3=8456*(19/2000).^2*pi*618.35*9.8/1000*ones(1,k);

DD=(D-G1-G2-G3)*1000;

% U=U*4.2/(max(U)-min(U));

av_n=10; %付氏级数所取项数，本文取 10

%以下为四个付氏系数

dert_o=zeros(1,av_n);

tao_t=zeros(1,av_n);

```

v=zeros(1,av_n);
q=zeros(1,av_n);

%以下为四个特解系数
aifa_a=zeros(1,av_n);
beita_b=zeros(1,av_n);
k_k=zeros(1,av_n);
miu_u=zeros(1,av_n);

%以下为载荷和位移值的一部分
F1=zeros(1,k);
U1=zeros(1,k);
F2=zeros(1,k);
U2=zeros(1,k);
F3=zeros(1,k);
U3=zeros(1,k);

%以下为参数
E=2.1e11;           %E 为钢杆弹性模量 Dt=70;
Dt=44;              %Dt 油管内径 mm
D1=25;              %D1 一级抽油杆直径 mm
D2=22;              %D2 二级抽油杆直径 mm
D3=19;              %D3 三级抽油杆直径 mm
A1=(D1/2000).^2*pi; %A1 一级抽油杆截面积 m.^2
A2=(D2/2000).^2*pi; %A2 二级抽油杆截面积 m.^2
A3=(D3/2000).^2*pi; %A3 三级抽油杆截面积 m.^2
u=3e7;              %u 液体粘度, Pa.s
L1=523.61;          %L1 一级抽油杆长度 m
L2=664.32;          %L2 二级抽油杆长度 m
L3=618.35;          %L3 三级抽油杆长度 m
pr=8456;             %pr 抽油杆密度 kg/m.^3
w=pi/7.5;           %w 为曲柄的角速度
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
a=(E/pr).^(1/2);      %a 为应力波在抽油杆柱中的传播速度
x1=523.61;
x2=664.32;
x3=618.35;
%x=[523.61,664.32,618.35];

%以下为求阻尼系数
m=Dt/D1;
B1=(m.^2-1)/2/log(m)-1;
B2=m.^4-1-(m.^2-1).^2/log(m);
%C=2*pi*u/pr/A1*(1/log(m)+2/B2*(B1+1)*(B1+2/((w*L1/a)/sin(w*L1/a)+cos(w*L1/a))));

```

```

C=0.1;
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%5
%以下为一级抽油杆泵功图
for n=1:av_n
    for p=1:k
        %以下为四个付氏系数
        dert_o(n)=dert_o(n)+2/k*DD(p)*cos(2*n*pi*p/k);% n=0,1,2,3...,av_n
        tao_t(n)=tao_t(n)+2/k*DD(p)*sin(2*n*pi*p/k); % n=1,2,3...,av_n
        v(n)=v(n)+2/k*U(p)*cos(2*n*pi*p/k);% n=0,1,2,3...,av_n
        q(n)=q(n)+2/k*U(p)*sin(2*n*pi*p/k);% n=1,2,3...,av_n
    end
    %以下为四个特解系数
    aifa_a(n)=n*w/a/sqrt(2)*sqrt(1+sqrt(1+(C/n/w).^2)); %C 为阻尼系数%a 为应力波在在抽油杆柱中的传播速度
    beita_b(n)=n*w/a/sqrt(2)*sqrt(-1+sqrt(1+(C/n/w).^2)); %
    k_k(n)=(dert_o(n).*aifa_a(n)+tao_t(n).*beita_b(n))/(E*A1*(aifa_a(n).^2+beita_b(n).^2));%E 为钢杆弹性模量, A1 为抽油杆柱的截面积, w 为曲柄的角速度
    miu_u(n)=(dert_o(n).*beita_b(n)+tao_t(n).*aifa_a(n))/(E*A1*(aifa_a(n).^2+beita_b(n).^2));
    %以下为特殊函数

    O(n)=(k_k(n)*cosh(beita_b(n)*x1)+q(n)*sinh(beita_b(n)*x1))*sin(aifa_a(n)*x1)+(miu_u(n)*sinh(beita_b(n)*x1)+v(n)*cosh(beita_b(n)*x1))*cos(aifa_a(n)*x1);

    P(n)=(k_k(n)*sinh(beita_b(n)*x1)+q(n)*cosh(beita_b(n)*x1))*cos(aifa_a(n)*x1)-(miu_u(n)*cosh(beita_b(n)*x1)+v(n)*sinh(beita_b(n)*x1))*sin(aifa_a(n)*x1);

    dO(n)=(tao_t(n)/E/A1*sinh(beita_b(n)*x1)+(q(n)*beita_b(n)-v(n)*aifa_a(n))*cosh(beita_b(n)*x1))*sin(aifa_a(n)*x1)+(dert_o(n)/E/A1*cosh(beita_b(n)*x1)+(v(n)*beita_b(n)+q(n)*aifa_a(n))*sinh(beita_b(n)*x1))*cos(aifa_a(n)*x1);

    dP(n)=(tao_t(n)/E/A1*cosh(beita_b(n)*x1)+(q(n)*beita_b(n)-v(n)*aifa_a(n))*sinh(beita_b(n)*x1))*cos(aifa_a(n)*x1)-(dert_o(n)/E/A1*sinh(beita_b(n)*x1)+(v(n)*beita_b(n)+q(n)*aifa_a(n))*cosh(beita_b(n)*x1))*sin(aifa_a(n)*x1);
end
dert_o_0=2/k*sum(DD);
v0=2/k*sum(U);
for N=1:k
    t=(60/4/142)*(N-1);
    for n=1:10
        U1(N)=U1(N)+O(n)*cos(n*w*t)+P(n)*sin(n*w*t);
        F1(N)=F1(N)+dO(n)*cos(n*w*t)+dP(n)*sin(n*w*t);
    end
    Uxt(N)=dert_o_0*x1/2/E/A1+v0/2+U1(N)
    Fxt(N)=(dert_o_0/2+E*A1*F1(N))/1000
end

```

%%%

%以下求二级杆的泵功图

v0_2=dert_o_0*x1/E/A1+v0;

v_2=0;%O wei hanshu bu wei ling

q_2=P;

dert_o_0_2=dert_o_0;

dert_o_2=E*A1*dO;

tao_t_2=E*A1*dP;

%以下为四个特解系数

for n=1:av_n

 aifa_a_2(n)=n*w/a/sqrt(2)*sqrt(1+sqrt(1+(C/n/w).^2)); %C 为阻尼系数%a 为应力波在在抽油杆柱中的传播速度

 beita_b_2(n)=n*w/a/sqrt(2)*sqrt(-1+sqrt(1+(C/n/w).^2)); %

 k_k_2(n)=(dert_o_2(n).*aifa_a_2(n)+tao_t_2(n).*beita_b_2(n))/(E*A2*(aifa_a_2(n).^2+beita_b_2(n).^2));%E 为钢杆弹性模量，A1 为抽油杆柱的截面积，w 为曲柄的角速度

 miu_u_2(n)=(dert_o_2(n).*beita_b_2(n)+tao_t_2(n).*aifa_a_2(n))/(E*A2*(aifa_a_2(n).^2+beita_b_2(n).^2));

 %以下为特殊函数

O_2(n)=(k_k_2(n)*cosh(beita_b_2(n)*x2)+q_2(n)*sinh(beita_b_2(n)*x2))*sin(aifa_a_2(n)*x2)+(miu_u_2(n)*sinh(beita_b_2(n)*x2)+v_2(n)*cosh(beita_b_2(n)*x2))*cos(aifa_a_2(n)*x2);

P_2(n)=(k_k_2(n)*sinh(beita_b_2(n)*x2)+q_2(n)*cosh(beita_b_2(n)*x2))*cos(aifa_a_2(n)*x2)-(miu_u_2(n)*cosh(beita_b_2(n)*x2)+v_2(n)*sinh(beita_b_2(n)*x2))*sin(aifa_a_2(n)*x2);

dO_2(n)=(tao_t_2(n)/E/A2*sinh(beita_b_2(n)*x2)+(q_2(n)*beita_b_2(n)-v_2(n)*aifa_a_2(n))*cosh(beita_b_2(n)*x2))*sin(aifa_a_2(n)*x2)+(dert_o_2(n)/E/A2*cosh(beita_b_2(n)*x2)+(v_2(n)*beita_b_2(n)+q_2(n)*aifa_a_2(n))*sinh(beita_b_2(n)*x2))*cos(aifa_a_2(n)*x2);

dP_2(n)=(tao_t_2(n)/E/A2*cosh(beita_b_2(n)*x2)+(q_2(n)*beita_b_2(n)-v_2(n)*aifa_a_2(n))*sinh(beita_b_2(n)*x2))*cos(aifa_a_2(n)*x2)-(dert_o_2(n)/E/A2*sinh(beita_b_2(n)*x2)+(v_2(n)*beita_b_2(n)+q_2(n)*aifa_a_2(n))*cosh(beita_b_2(n)*x2))*sin(aifa_a_2(n)*x2);

end

for N=1:k

 t=(60/4/142)*(N-1);

 for n=1:10

 U2(N)=U2(N)+O_2(n)*cos(n*w*t)+P_2(n)*sin(n*w*t);

 F2(N)=F2(N)+dO_2(n)*cos(n*w*t)+dP_2(n)*sin(n*w*t);

 end

 Uxt_2(N)=dert_o_0_2*x2/2/E/A2+v0_2/2+U2(N)

 Fxt_2(N)=(dert_o_0_2/2+E*A2*F2(N))/1000

end

%%%

%以下求三级杆的泵功图

v0_3=dert_o_0_2*x2/E/A2+v0_2;

```

v_3=O_2;%O wei hanshu bu wei ling
q_3=P_2;
dert_o_0_3=dert_o_0_2;
dert_o_3=E*A2*dO_2;
tao_t_3=E*A2*dP_2;
%以下为四个特解系数
for n=1:av_n
    aifa_a_3(n)=n*w/a/sqrt(2)*sqrt(1+sqrt(1+(C/n/w).^2)); %C 为阻尼系数%a 为应力波在抽油杆柱中的传播速度
    beita_b_3(n)=n*w/a/sqrt(2)*sqrt(-1+sqrt(1+(C/n/w).^2)); %
    k_k_3(n)=(dert_o_3(n).*aifa_a_3(n)+tao_t_3(n).*beita_b_3(n))/(E*A3*(aifa_a_3(n).^2+beita_b_3(n).^2));%E
    miu_u_3(n)=(dert_o_3(n).*beita_b_3(n)+tao_t_3(n).*aifa_a_3(n))/(E*A3*(aifa_a_3(n).^2+beita_b_3(n).^2));
    %以下为特殊函数

O_3(n)=(k_k_3(n)*cosh(beita_b_3(n)*x3)+q_3(n)*sinh(beita_b_3(n)*x3))*sin(aifa_a_3(n)*x3)+(miu_u_3(n)*sinh(
beita_b_3(n)*x3)+v_3(n)*cosh(beita_b_3(n)*x3))*cos(aifa_a_3(n)*x3);

P_3(n)=(k_k_3(n)*sinh(beita_b_3(n)*x3)+q_3(n)*cosh(beita_b_3(n)*x3))*cos(aifa_a_3(n)*x3)-(miu_u_3(n)*cosh(
beita_b_3(n)*x3)+v_3(n)*sinh(beita_b_3(n)*x3))*sin(aifa_a_3(n)*x3);

dO_3(n)=(tao_t_3(n)/E/A3*sinh(beita_b_3(n)*x3)+(q_3(n)*beita_b_3(n)-v_3(n)*aifa_a_3(n))*cosh(beita_b_3(n)*
x3))*sin(aifa_a_3(n)*x3)+(dert_o_3(n)/E/A3*cosh(beita_b_3(n)*x3)+(v_3(n)*beita_b_3(n)+q_3(n)*aifa_a_3(n))*s
inh(beita_b_3(n)*x3))*cos(aifa_a_3(n)*x3);

dP_3(n)=(tao_t_3(n)/E/A3*cosh(beita_b_3(n)*x3)+(q_3(n)*beita_b_3(n)-v_3(n)*aifa_a_3(n))*sinh(beita_b_3(n)*
x3))*cos(aifa_a_3(n)*x3)-(dert_o_3(n)/E/A3*sinh(beita_b_3(n)*x3)+(v_3(n)*beita_b_3(n)+q_3(n)*aifa_a_3(n))*c
osh(beita_b_3(n)*x3))*sin(aifa_a_3(n)*x3);
end
for N=1:k
    t=(60/4/142)*(N-1);
    for n=1:10
        U3(N)=U3(N)+O_3(n)*cos(n*w*t)+P_3(n)*sin(n*w*t);
        F3(N)=F3(N)+dO_3(n)*cos(n*w*t)+dP_3(n)*sin(n*w*t);
    end
    Uxt_3(N)=dert_o_0_3*x3/2/E/A3+v0_3/2+U3(N)
    Fxt_3(N)=(dert_o_0_3/2+E*A3*F3(N))/1000
end
N=0:(60/4/142):(60/4);
figure(1);
plot(Uxt,Fxt,'r');
hold on;
plot(Uxt_2,Fxt_2,'b');
hold on;

```

```

plot(Uxt_3,Fxt_3,'g');
hold on;
plot(U,D,'p');

figure(2)
subplot(2,2,1);plot(N,U);
subplot(2,2,2);plot(N,D);
subplot(2,2,3);plot(N,Uxt);
subplot(2,2,4);plot(N,Fxt);
figure(3);
subplot(2,2,1);plot(N,Uxt_2);
subplot(2,2,2);plot(N,Fxt_2);
subplot(2,2,3);plot(N,Uxt_3);
subplot(2,2,4);plot(N,Fxt_3);
%以下为求有效冲程
%求曲率
for i=2:k-1
    P(i)=sqrt((Fxt_3(i)-Fxt_3(i-1)).^2+(Uxt_3(i)-Uxt_3(i-1)).^2);
    P(i-1)=sqrt((Fxt_3(i-1)-Fxt_3(i+1)).^2+(Uxt_3(i-1)-Uxt_3(i+1)).^2);
    P(i+1)=sqrt((Fxt_3(i+1)-Fxt_3(i)).^2+(Uxt_3(i+1)-Uxt_3(i)).^2);
    PP(i)=(P(i)+P(i-1)+P(i+1))/2;
    S(i)=sqrt(PP(i)*(PP(i)-P(i))*(PP(i)-P(i-1))*(PP(i)-P(i+1))));
    K(i)=2*S(i)/P(i)/P(i-1)/P(i+1);
end
%求曲率变化量
len_K=length(K); %143
deto_K=zeros(1,len_K-1);
for i=1:len_K-1
    deto_K(i)=abs(K(i+1)-K(i));
end
%精确曲率变化量
len_deto_K=length(deto_K); %142
av_deto_K=zeros(1,len_deto_K-4);
for i=3:len_deto_K-2
    av_deto_K(i)=(deto_K(i-2)+deto_K(i-1)+deto_K(i)+deto_K(i+1)+deto_K(i+2))/5;
end
%求变化量最大点
IndMin=find(diff(sign(diff(av_deto_K)))>0)+1;
IndMax=find(diff(sign(diff(av_deto_K)))<0)+1;
figure(7);
plot(1:length(av_deto_K),av_deto_K);
hold on;
plot(IndMax,av_deto_K(IndMax),'k*')

```

N=[33,85,111,9];%四边形的右下角为起始点，逆时针转动，第一个点表示右下角，第二个点表示左下角，第三个左上角，第四个右上角。

```
U_point=zeros(1,4);
```

```
D_point=zeros(1,4);
```

```
for i=1:4
```

```
    n=N(i);
```

```
    U_point(i)=Uxt_3(n);
```

```
    D_point(i)=Fxt_3(n);
```

```
end
```

```
L_eff=Uxt(111)-Uxt(9)    %有效冲程
```

```
figure(5)
```

```
plot(U_point,D_point,'r');
```

```
hold on;
```

```
plot(Uxt_3,Fxt_3,'b');
```

```
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
```

```
%以下为有效面积法求抽油量
```

```
%以下为泵功图面积
```

```
S_beng=0;
```

```
S_li=0;
```

```
for i=1:142
```

```
    width=Uxt(i+1)-Uxt(i);
```

```
    line=(Fxt(i+1)+Fxt(i))/2;
```

```
    S_beng=S_beng+width*line;
```

```
end
```

```
S_li=(max(D)-min(D))*L_eff;
```

```
%有效面积法计算抽油量
```

```
qp=1440*pi*(44/1000).^2*4.2*4*0.969/4
```

```
K=S_beng/S_li
```

```
q_area=K*qp
```

```
%有效冲程计算抽油量
```

```
q_eff=1440*pi*(44/1000).^2*4*988.9*L_eff/4/1000    %换算成吨
```